

Zaid Ahmad

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Education

B.Sc. Computer Science, AI and Machine Learning Stream | *Carleton University* Ottawa, ON Sept 2024 – May 2028

Coursework: Data Structures and Algorithms, C/C++, Linear Algebra, Discrete Structures II, Web Applications

Technical Skills

Languages: C++, C, Java, Python, JavaScript, TypeScript

Frameworks and Libraries: React (Next.js), React Flow, Node.js, Express, Tailwind CSS, Framer Motion

Tools and Platforms: Git/GitHub, GitHub Actions, Firebase, MongoDB, Supabase, Linux, Netlify, Render, POSIX Threads, Microsoft Office (Word, Excel, PowerPoint)

DevOps and Cloud: FastAPI, PostgreSQL, Docker, Vercel, REST APIs, Linux (Ubuntu/Mint)

Experience

Project Officer (FSWEP) | *Immigration, Refugees and Citizenship Canada*

Ottawa, ON June 2026 – Aug 2026

- Reduced analyst hours spent on repetitive data entry by building Python automation scripts that parsed and routed IT intake requests, freeing capacity for higher-priority triage work.
- Improved requirements documentation quality for 10+ intake requests by translating client needs into structured business requirements, reducing back-and-forth with stakeholders before sign-off.
- Accelerated senior management reporting cycles by consolidating intake data from multiple sources into a standardized reporting format, cutting ad-hoc information requests to the team.
- Increased CIIT correspondence throughput by drafting and editing team documentation and stakeholder communications, decreasing revision rounds before senior-level approval.

Logistics Coordinator | *cuHacking*

Ottawa, ON Sept 2025 – Present

- Reduced day-of coordination failures at a 200+ participant hackathon by authoring pre-event checklists and run-of-show documents covering scheduling, room assignments, and AV equipment.
- Prevented resource shortfalls across 10+ teams by auditing buffer stock before the event and flagging inventory gaps early, eliminating last-minute procurement blockers on event day.

Projects

CarletonCourseMap | carletoncoursemap.ca | github.com/zaidahmad16/CarletonCourseMap

Python - Parsel/XPath - FastAPI - PostgreSQL - Next.js - React Flow - dagre - Tailwind CSS - Railway - Cloudflare

- Eliminated manual degree planning for 700+ Carleton students by scraping Carleton Central with Parsel/XPath, modelling AND/OR prerequisite logic as a DAG, and storing course data as JSONB in PostgreSQL on Neon.
- Surfaced prerequisite conflicts in real time by rendering four-year course dependency graphs with React Flow and dagre, with color-coded nodes and monthly-refreshed professor ratings from RateMyProfessors.
- Reduced deployment friction by shipping the Next.js frontend and FastAPI backend on Railway under a custom domain managed via Cloudflare.

STM32 Doom Bootloader | github.com/zaidahmad16/stm32-doom-loader

C - ARM Cortex-M4 - Linker Scripts - OpenOCD - STM32F4

- Achieved reliable bare-metal application launch by writing a stage-1 bootloader at 0x08000000 with CRC-32 image verification, halting on corruption and jumping to a Doom application at 0x08010000 on success.
- Enabled full graphical boot with no OS by writing bare-metal SPI and SD drivers for an ILI9341 LCD without HAL dependencies, rendering a splash screen and loading the shareware WAD from SD card.

Linux Wireless Kernel Patch | github.com/zaidahmad16/mac80211-ht-downgrade-fix

C - Linux Kernel - Wireless Networking

- Restored wireless throughput from 54 Mbps to full HT rates by diagnosing a firmware incompatibility between a WiFi adapter and router beacon, then patching the mac80211 driver directly in C.

Multithreaded C Simulation | github.com/zaidahmad16/COMP2401-Final-Project

C - POSIX Threads - Semaphores

- Achieved stable execution across 1,000+ simulated ticks with zero deadlocks or memory leaks by implementing a concurrent agent simulation using POSIX threads, semaphore synchronization, and manual memory management.

Certifications and Clearances

Government of Canada Reliability Status Security Clearance – Issued 2026